V06: physical failure simulation for a 500 m³ pool

Executed Simulink R2026a | 7 cases, 168 h each | synthetic inputs

1. Software shutdown versus physical leakage

The new standalone model is derived from v05. Its two simulated temperature channels have separate 15/30-minute delays. A valve stuck open can physically deliver heat despite zero pool heating command. Insufficient independent data-centre heat rejection leaves unhandled cooling heat. Original v85, v04 and v05 are unchanged. No facility measurements were used.

2. Energy equations and conservation

Uncontrolled heat transfers out of existing commanded reject heat, so no energy is invented. The lumped IT heat buffer is merely a scenario risk indicator, NOT a measured server or coolant temperature.

C_pool = 997 * 500 * 4182 = 2 084 727 000 J/K

T_pool(k+1) = T_pool(k) + 900 (Q_pool_physical - Q_loss)/C_pool

T_sensor1(k) = T_true(k-1) + bias1; T_sensor2(k) = T_true(k-2) + bias2

Q_leak = min(25 kW, Q_reject_command, 4.5 kW/K * max(0, T_DC - T_pool - 3 K))

Q_pool_physical = Q_pool_command + Q_leak

Q_reject_physical = min(Q_reject_command - Q_leak, 180 kW * availability)

Q_unhandled = Q_reject_command - Q_leak - Q_reject_physical

Q_DC = Q_HX_physical + Q_HP_source + Q_reject_physical + Q_unhandled

C_IT = 8e6 J/K; T_IT(k+1) = max(30, T_IT(k) + 900*(Q_unhandled - Q_passive)/C_IT)

Q_passive = max(0, 2000 W/K * (T_IT - 30 C)); IT risk alarm at 42 C (scenario).

3. Safety-state logic and limitations

The simulated sensors have no access to the true current water temperature. A disagreement above 0.35 K or measured temperature above 27.70°C trips a latched shutdown. Manual reset requires cleared faults and readings at or below 27.65°C. Identical bias on both sensors was NOT detected, illustrating a common-mode risk.

4. Actual completed scenario outputs

Scenario	Leak MWh	Unrej. MWh	IT max °C	Latch
Healthy	0.000	0.000	30.00	0
One sensor bias	0.000	0.000	30.00	48
Common bias	0.000	0.000	30.00	0
Valve stuck	0.490	0.000	30.00	108
REJECT down	0.000	0.080	47.06	0
No pool flow	0.000	0.000	30.00	52
Combined failure	0.153	0.057	42.72	64

5. Independent verification and gaps

84,801 independent checks passed on exported Simulink CSV, including pool and DC energy balances, sensor delays and the latch. A passing model is not a physical safety certificate. Real sensor independence, redundant isolation valves, documented reject capacity, hydraulic design, measured COP maps and measured annual loads remain essential.

QA: 84801 PASS; 7 * 672 time steps (dt=900 s).

Provenance D: synthetic. Model buffer IT temperature is NOT an actual server temperature.

EU 2024/1364: https://eur-lex.europa.eu/eli/reg_del/2024/1364/oj

6. Actual simulated time series



